Penn Engineering
University of Pennsylvania

FA26

Special Seminar on
Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

https://cis7000-fall26.thelabone.org/

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Lecture 11: Video Generation and Prediction

Special Seminar on Multi-Modal AI

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~~Policies~~ Ideas for Writing with AI

- You own your words
- Make sure you have read **every single sentence** you submitted
- In particular, watch out for “terminologies” they created
  - Within each paper, the name for a particular concept should be consistent
  - e.g., residuals, noises, errors, …
- Feel free to use AI to help you explain things, but make sure you write the statement for each paragraph

General Feedbacks

- Be sure to include data collection processes (or referring to existing datasets)
- Be sure to mention your evaluation metric (and discuss about it)
- AI can help you draft things; but turn on your brain's "Extra High" thinking mode

- If you have questions, reach out!

Lecture overview

1 Problem Setup

2 Video Representations

3 Controllable Generation

4 Advanced Inference Techniques

1. Problem Setups

Video Generation Tasks

Task	Observed input	Generated output
Text-to-video	A description	An entire video
Future prediction	Past frames, optionally actions	Later frames
Interpolation	Frames before and after a gap	Missing frames

The same scene can pose different conditional inference problems

$$p(x_G \mid x_H, c)$$

Song et al., History-Guided Video Diffusion (2025) · §3–4

Example: Video Prediction

A camera moves through a kitchen toward a doorway

We need plausible motion while preserving the room already observed



Song et al., History-Guided Video Diffusion (2025) · Fig. 1, cropped

Evaluation Metrics

Criterion	Evidence
Distributional realism	FVD in a video feature space
Prompt or task alignment	VBench dimensions, human review
Temporal consistency	Frame or feature consistency
Motion and diversity	Dynamic degree, diverse samples

Song et al., History-Guided Video Diffusion (2025) · §6; see also CogVideoX §4 and VideoGPT §4

2. Video Representations for Generation

Compression to Make Modeling Tractable

Videos are big:

$$16 \times 64 \times 64 \;=\; 65{,}536 \qquad 4 \times 16 \times 16 \;=\; 1{,}024$$

Downsampling each axis by four gives 64 times fewer sites

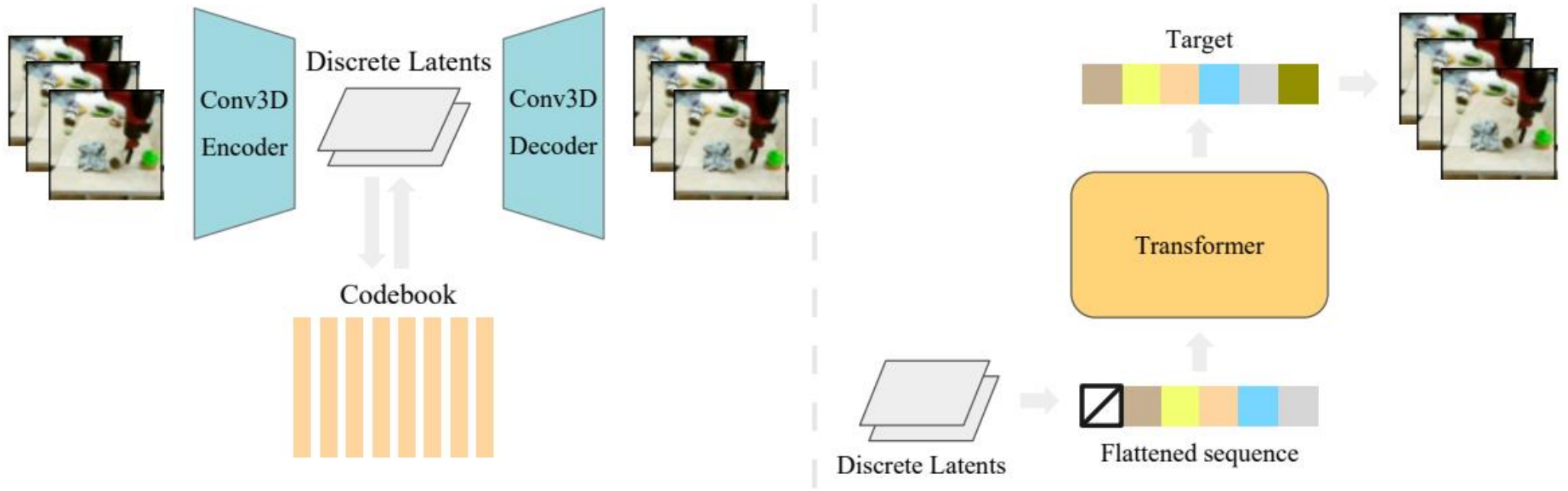
But a naïve dowmsampling is not great (information loss)

Yan et al., VideoGPT (2021) · §1 and §4.7

VideoGPT: VQ-VAE for AR Prediction

Key idea: turn video into discrete latent codes that a GPT can model

First learn reconstruction, then freeze the codec and train the generator



Yan et al., VideoGPT (2021) · Fig. 2; §2–3

VQ-VAE: Recap

Optional detail

Each encoder vector selects its nearest learned codebook entry

$$q = \arg\min_j \|E(x) - e_j\|^2$$

Reconstruction trains useful codes, and commitment stabilizes assignments

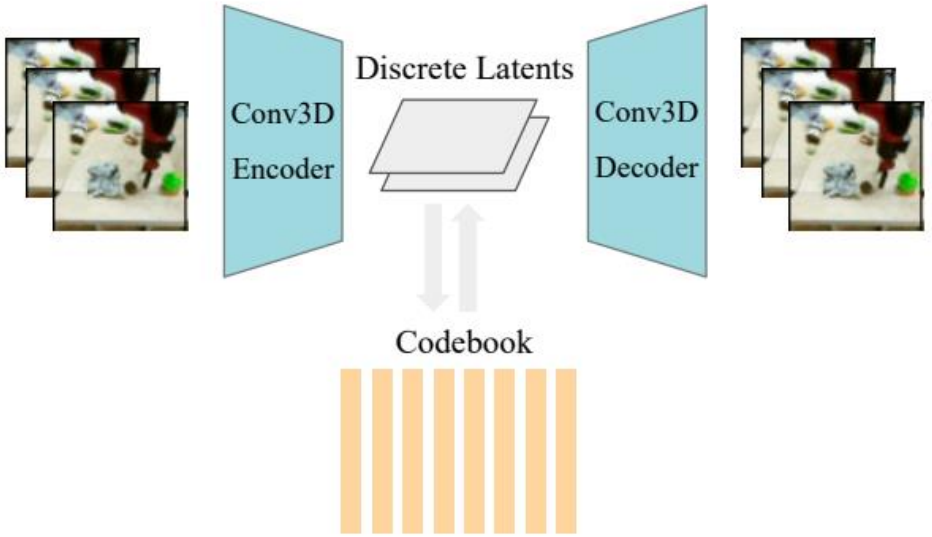
$$L = L_{recon} + \|sg[E(x)] - e\|^2 + \beta\|E(x) - sg[e]\|^2$$

Yan et al., VideoGPT (2021) · §2.1; §4.1

VQ-VAE: Recap

Optional detail

Each encoder vec ook entry



Reconstruction tı stabilizes assignments

$$L = L_{recon} + \|sg[E(x)] - e\|^2 + \beta\|E(x) - sg[e]\|^2$$

Yan et al., VideoGPT (2021) · §2.1; §4.1

Autoregressive Modeling of the Sequence

$$p(q \mid c) = \prod_{i=1}^{N} p_\theta(q_i \mid q_{<i}, c)$$

Flatten the spatiotemporal grid in a fixed order and use causal attention

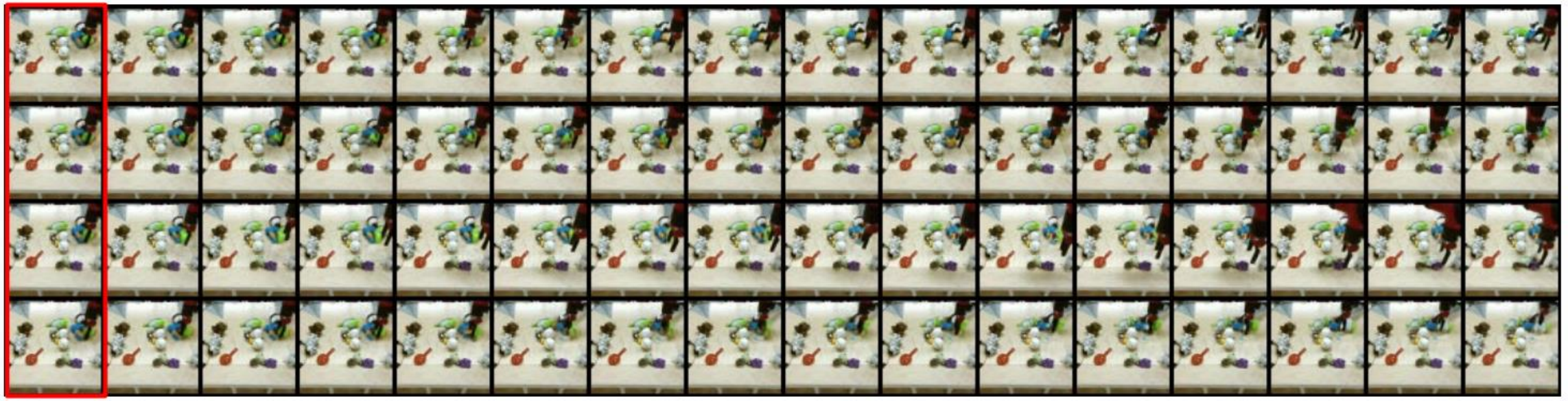
$$L_{prior} = -\sum_{i=1}^{N} \log p_\theta(q_i \mid q_{<i}, c)$$

Training uses ground-truth preceding codes (also called teacher-forcing); sampling uses generated codes

Yan et al., VideoGPT (2021) · §2.2–3

Prediction requires an explicit conditioning path

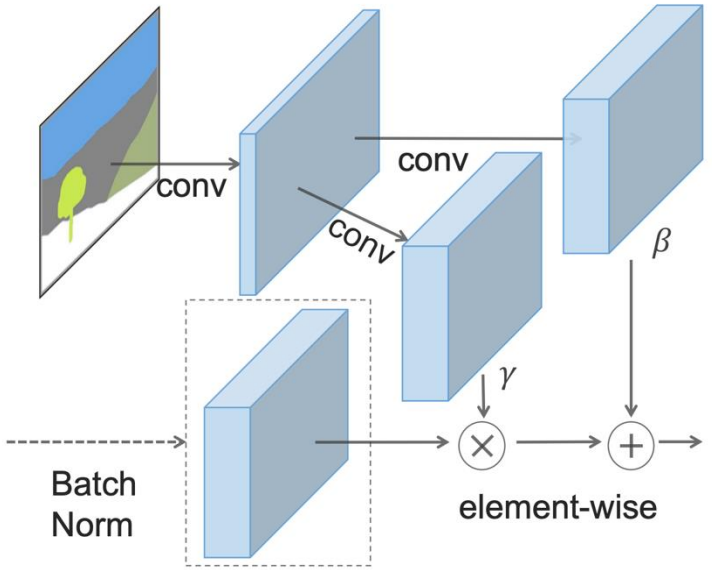
Class or action conditioning modulates Transformer LayerNorm parameters



Recall the idea from segmentation-to-image generation (SPADE)

Yan et al., VideoGPT (2021) · §3; Fig. 6, cropped

SPADE: Restore Layout after Normalization



- **Key idea:** Restore mask information after normalization using spatially varying scale and bias

Park, Liu, Wang & Zhu, CVPR 2019 · Fig. 2

VideoGPT: the representation sets the ceiling

Key idea: compress video first, then learn the distribution of its discrete codes

(Open) Questions:

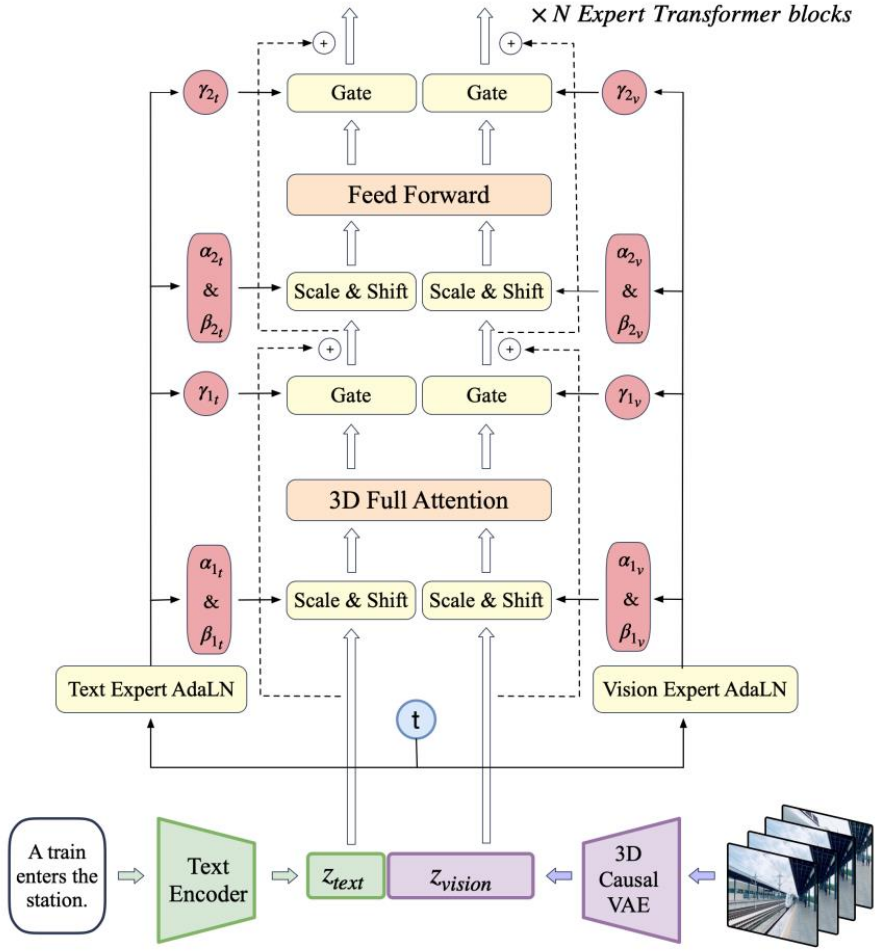
- The reconstruction-generation gap (more next week)
- The choice of AR modeling on an ordered sequence of the latents (more next lecture)

CogVideoX: Diffusion on VAE Latents

Key idea: jointly denoise compressed video using text and global spatiotemporal attention

A 3D causal VAE provides latents, and T5 provides text embeddings

The Transformer predicts a denoising target, then the VAE decodes



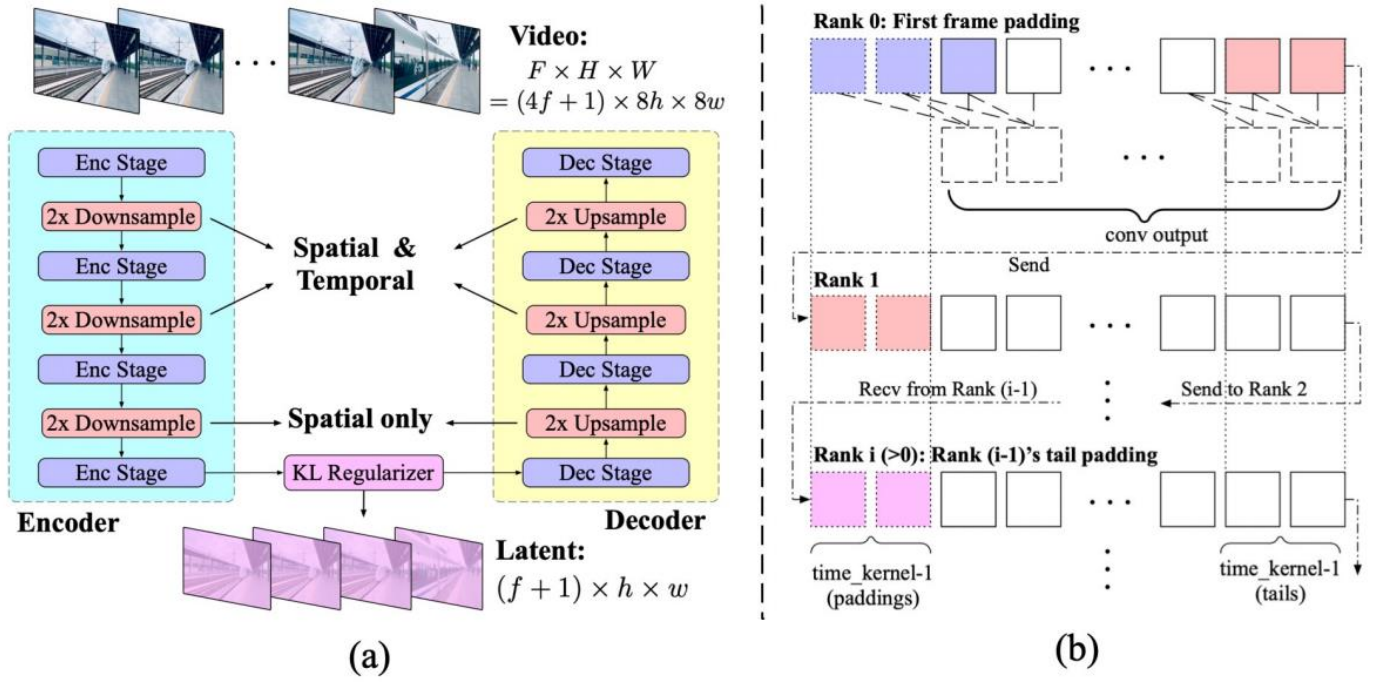
Yang et al., CogVideoX (2024), arXiv v1 · Fig. 2; §2–3

A Causal Video VAE

Optional detail

Temporal downsampling is 4× and spatial downsampling is 8×8

Causal convolutions keep later frames out of earlier latent features (Not obviously the best option)



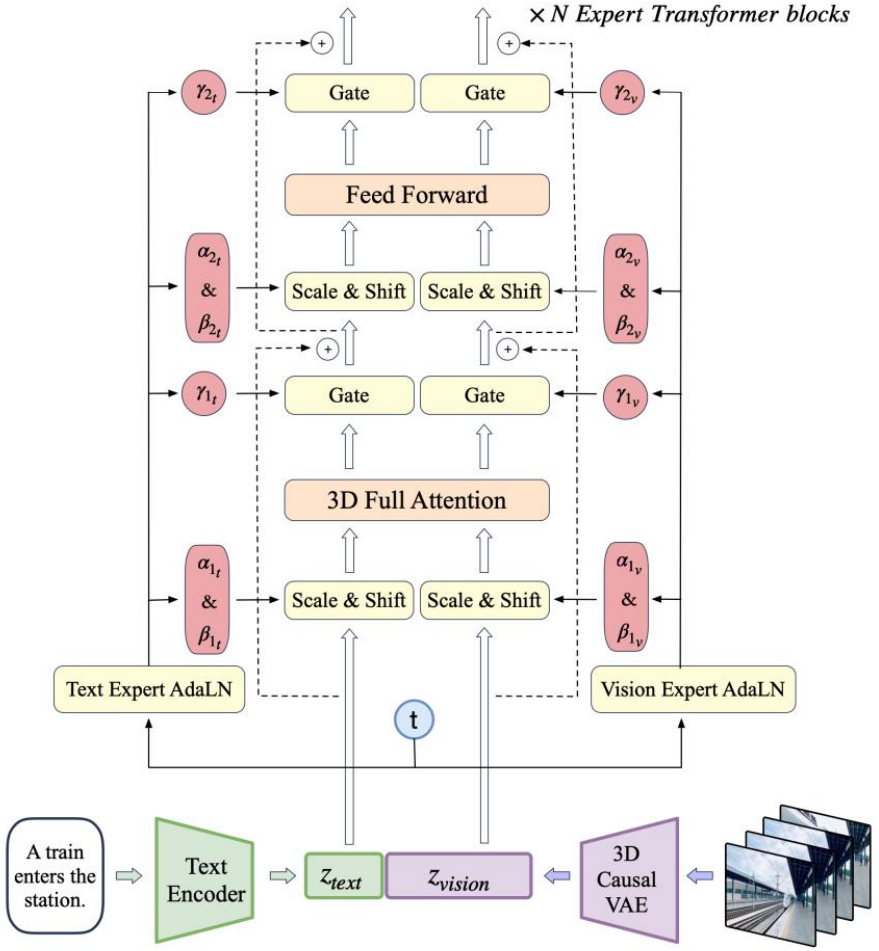
Yang et al., CogVideoX (2024), arXiv v1 · Fig. 3; §2.1

More on Normalization

Optional detail

Concatenate text and video tokens so attention can exchange information directly

Modality-specific adaptive LayerNorm handles different feature scales



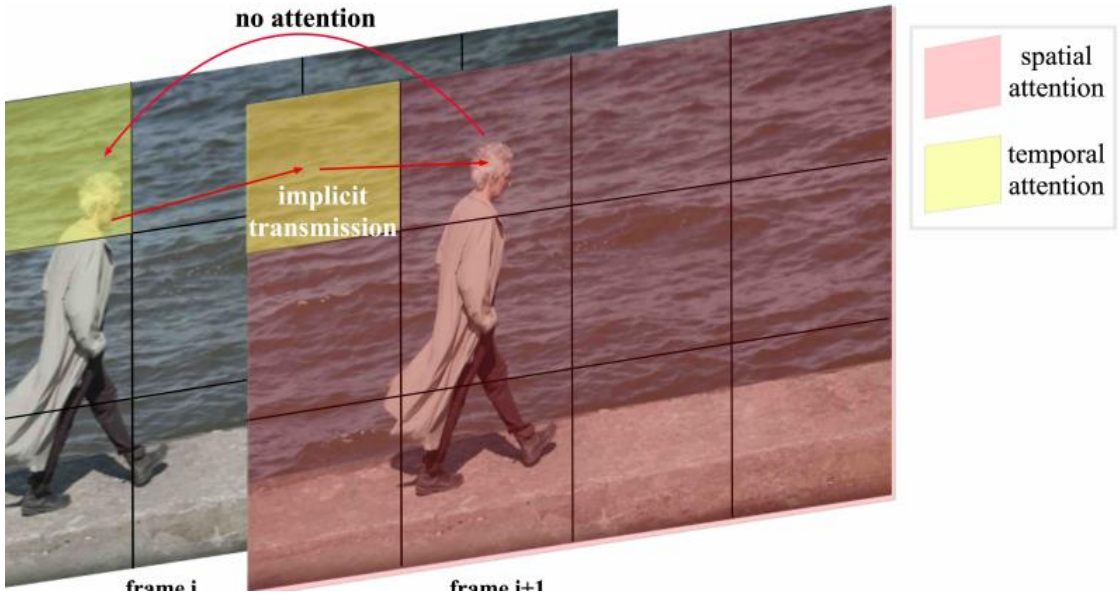
Yang et al., CogVideoX (2024), arXiv v1 · Fig. 2 detail; §2.2

Separated Spatial-Temporal Convolution is not Good

A moving object may occupy different patches in adjacent frames

Separated spatial and temporal attention routes information through intermediate patches

Solution: Full 3D attention



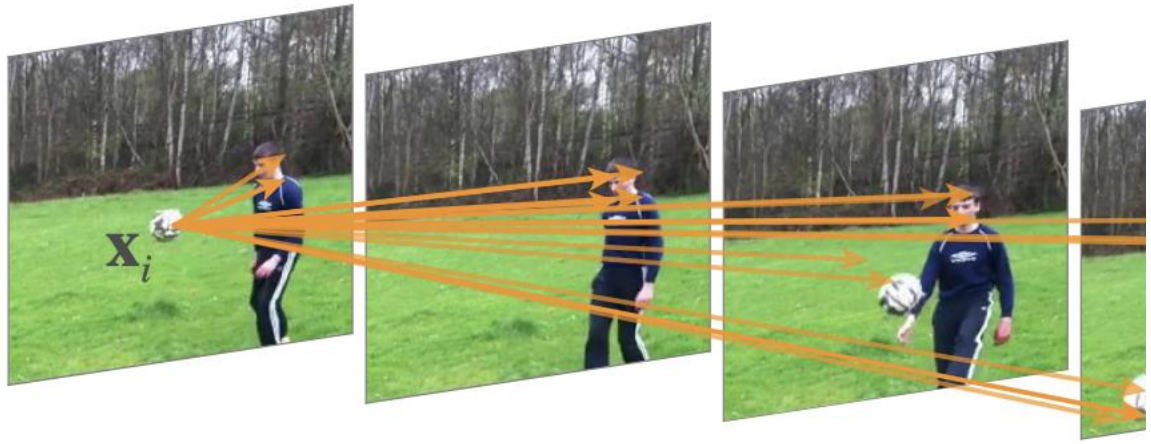
Recall the idea from video understanding (Non-Local Networks)

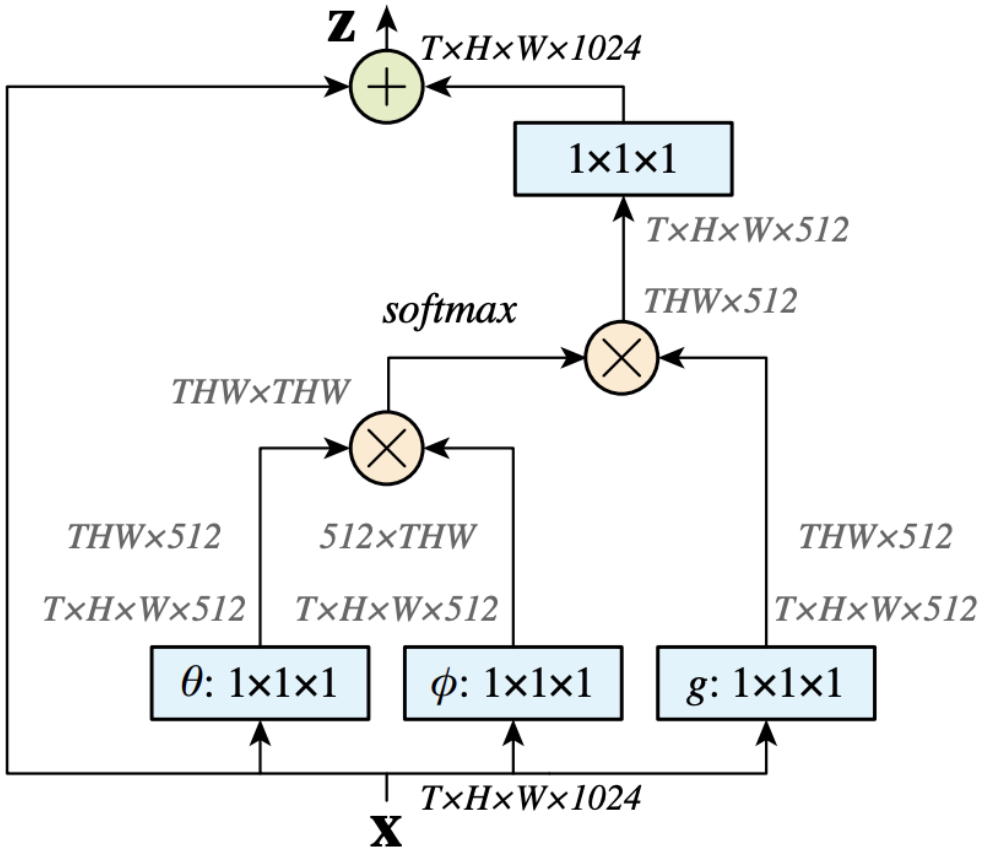
Yang et al., CogVideoX (2024), arXiv v1 · Fig. 5; §2.2

Two Ideas: Relational Features and Global Contexts

Non-local Neural Networks

Xiaolong Wang[1,2*] Ross Girshick[2] Abhinav Gupta[1]
[1]Carnegie Mellon University [2]Facebook AI Res





Motivating Task: Video Activity Classification

- We need to consider the feature of the same object in other frames

Training Target: Standard Diffusion

$$z^k = \alpha_k z + \sqrt{1 - a_k^2}\,\varepsilon$$

$$v = \alpha_k \varepsilon - \sqrt{1 - a_k^2}\,z \qquad L = E\,\|v_\theta(z^k, c, k) - v\|^2$$

The original implementation uses v-prediction

Yang et al., CogVideoX (2024), arXiv v1 · §3; standard v-prediction notation

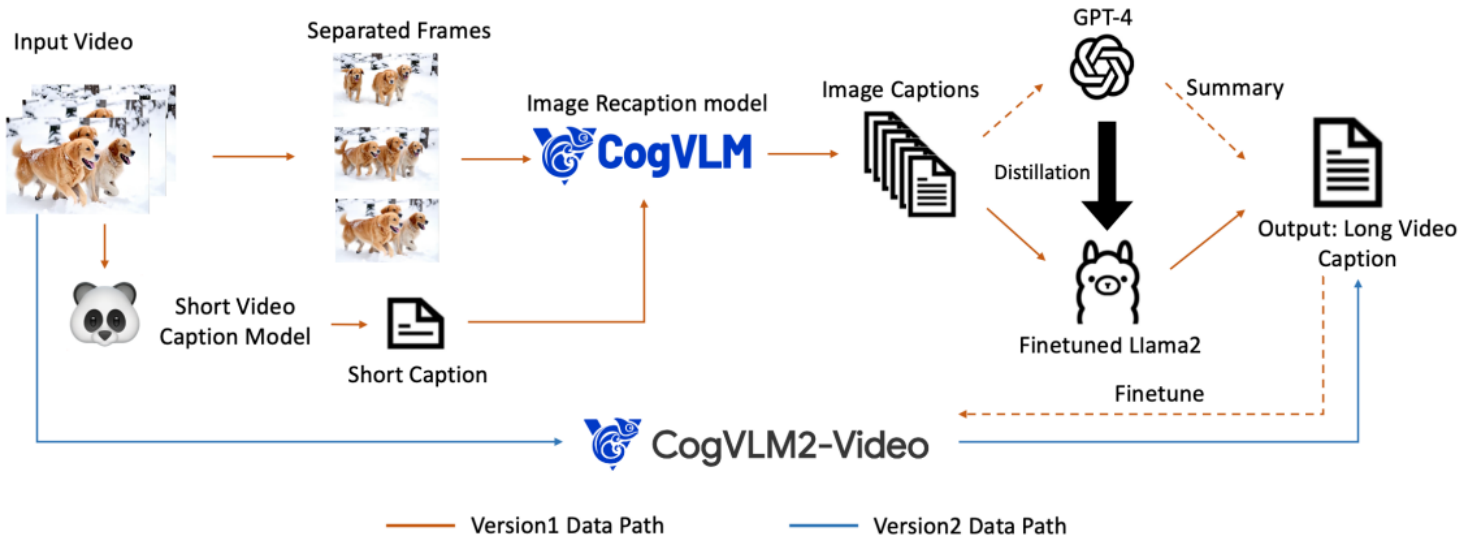
Data Generation

Optional detail

Filtering leaves about 35 million single-shot clips, averaging six seconds

Use both dense frame descriptions and a short video caption

Very standard strategy for large model training!



Yang et al., CogVideoX (2024), arXiv v1 · §3.4; Fig. 8

Scale and Data Quality

Optional detail

Model (2024 report)	Human action ↑	Multiple objects ↑	Dynamic quality ↑
CogVideoX-2B	88.0	53.70	57.7
CogVideoX-5B	96.8	70.95	69.5

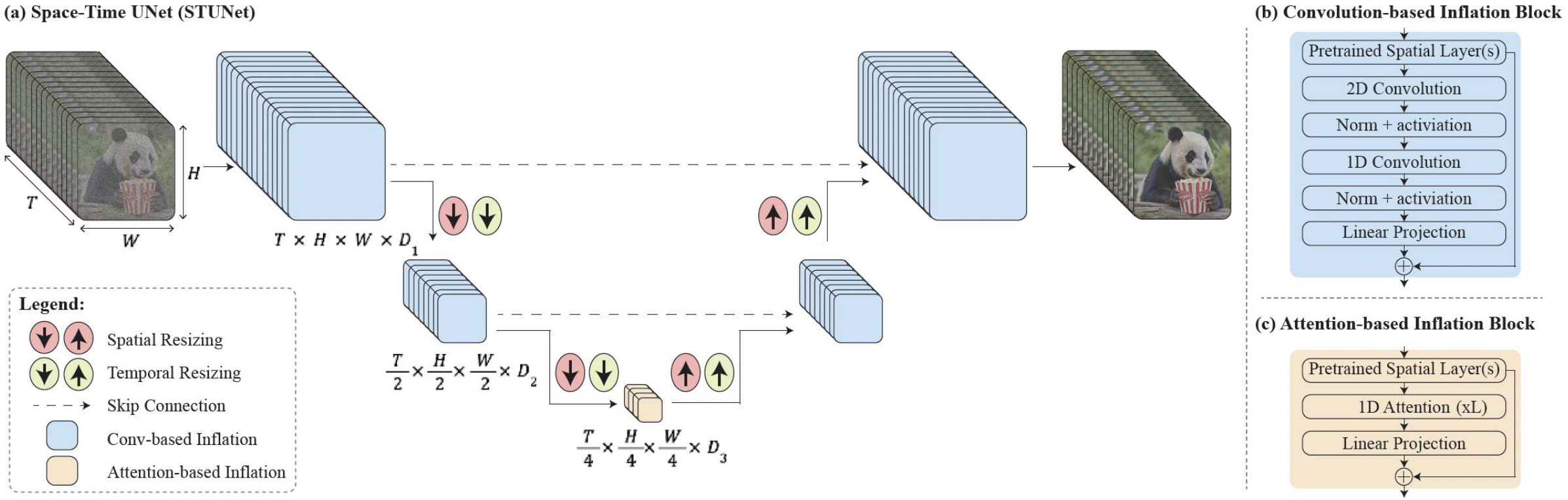
Different evaluation metrics

Yang et al., CogVideoX (2024), arXiv v1 · Table 1; §4

More Architecture Ideas: Resolution

Additional idea

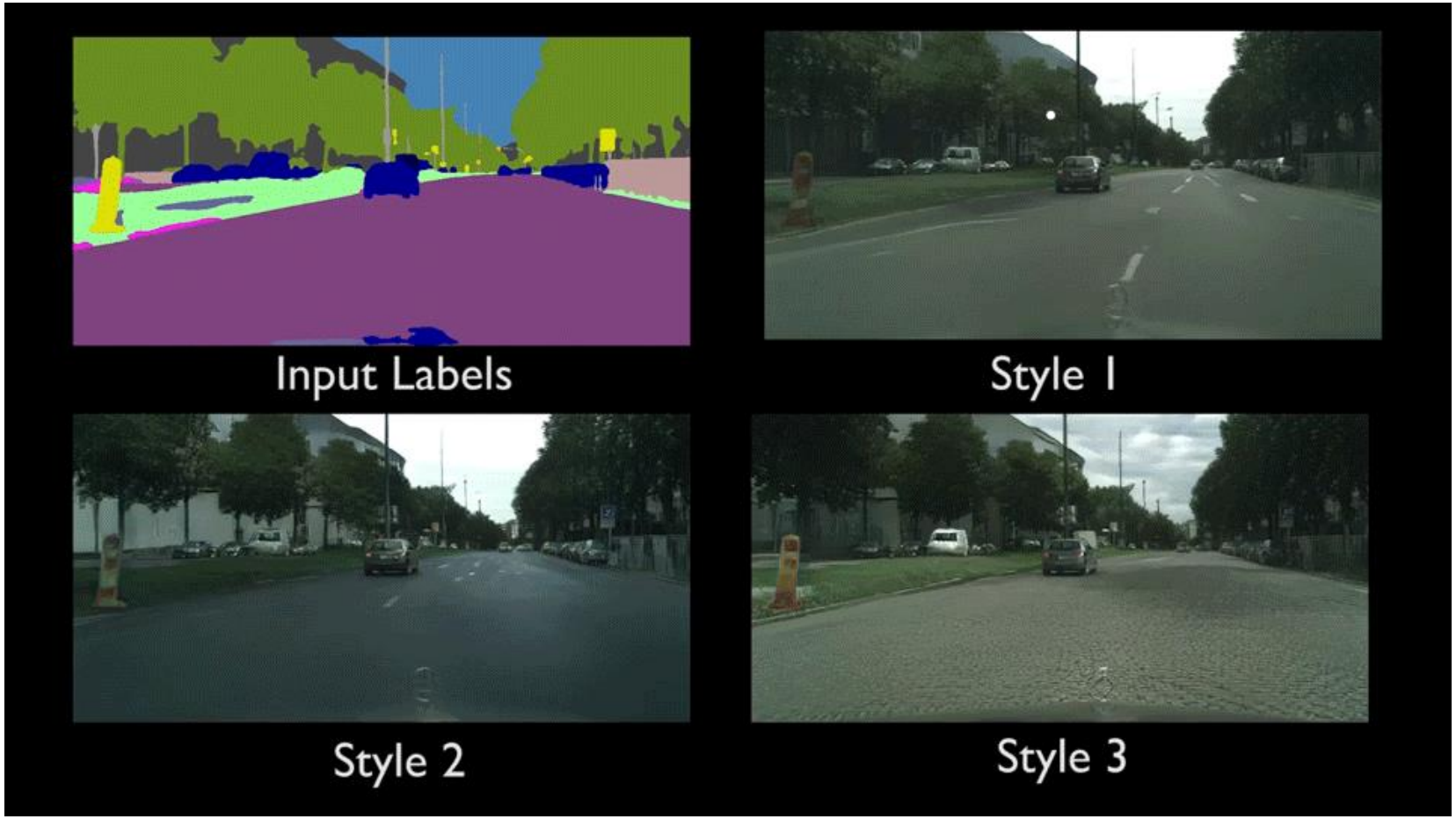
Key idea: use a spatio-temporal U-Net



Bar-Tal et al., Lumiere: A Space-Time Diffusion Model for Video Generation (2024), v2 · Fig. 3; §3

Leveraging Image Formation Ideas: Flow

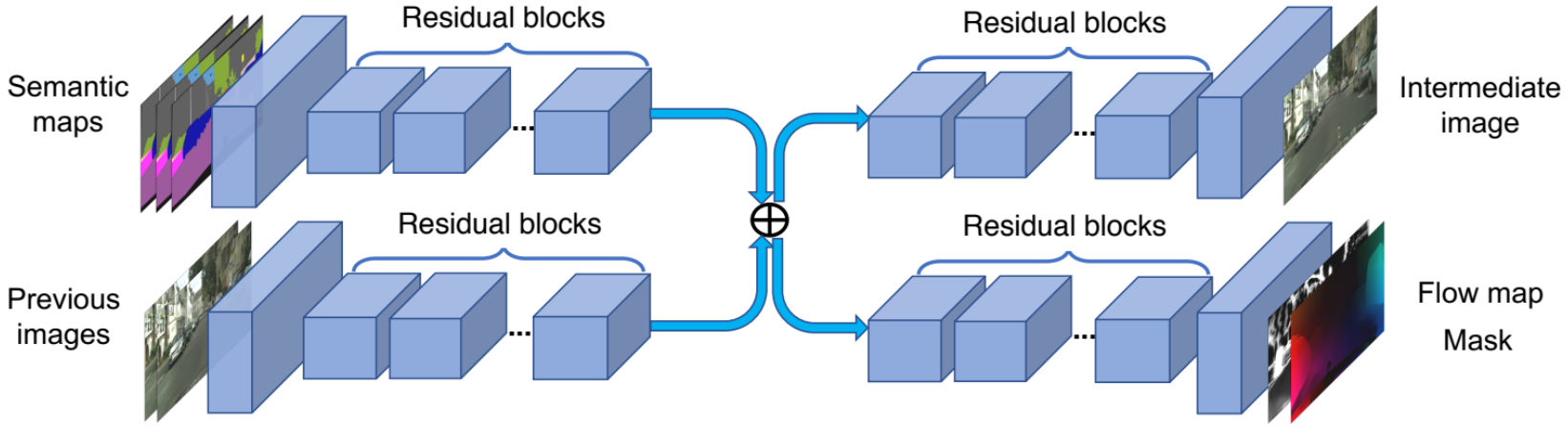
Application: Semantic-map conditioned video generation



Wang et al., Video-to-Video Synthesis (NeurIPS 2018) · Fig. 8; Eq. 4

Leveraging Image Formation Ideas: Flow

Key idea: predict flow to reuse past pixels, and a soft mask to decide newly synthesized pixels



$$y_t = (1 - m_t) \odot \text{Warp}(y_{t-1}, w_t) + m_t \odot h_t$$

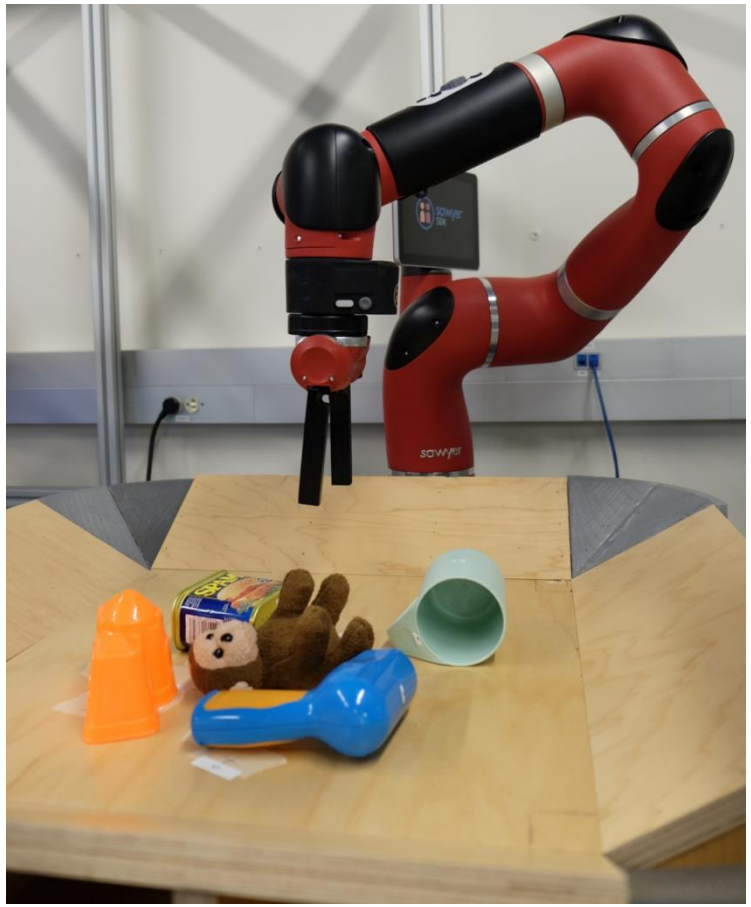
Wang et al., Video-to-Video Synthesis (NeurIPS 2018) · Fig. 8; Eq. 4

Leveraging Image Formation Ideas: Flow (for Memory)

Additional idea

Application: Learning a video for robot manipulation tasks

Motivation: Due to occlusions, we need to retrieve from a “memory” of previous frames

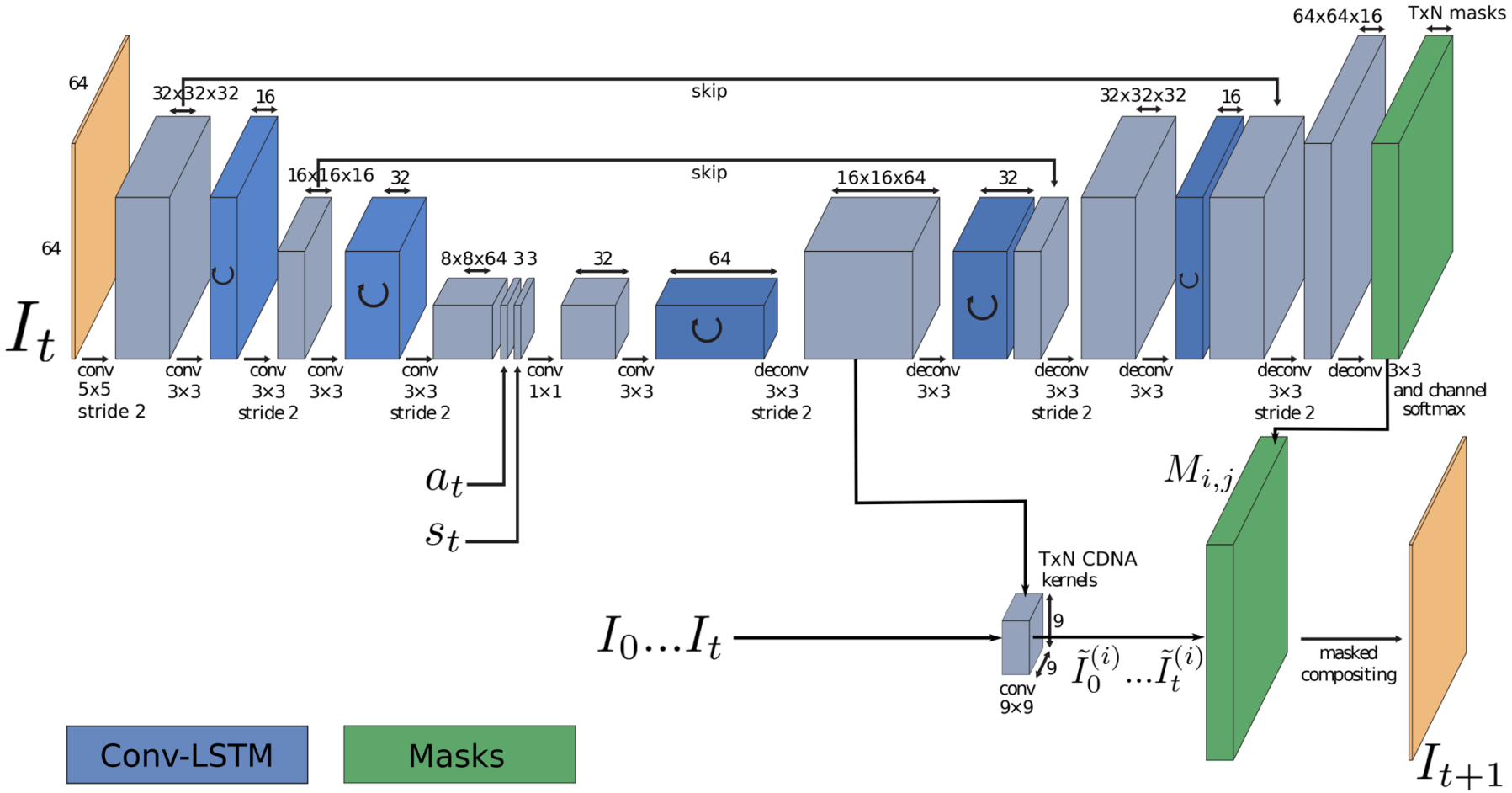


Ebert et al., Self-Supervised Visual Planning with Temporal Skip Connections (CoRL 2017) · Fig. 2; §4

Leveraging Image Formation Ideas: Flow (for Memory)

Additional idea

Key idea: an occluded object may need information from an older frame



Ebert et al., Self-Supervised Visual Planning with Temporal Skip Connections (CoRL 2017) · Fig. 2; §4

Summary: Ideas for Video Representations

- VQ-VAE or 3D VAE codecs to compress
- Spatiotemporal global attention
- Spatiotemporal resolution hierarchies
- Image formation Ideas: Flow

3 Controllable Generation

Explicit Control Signals

A prompt can describe a dance without specifying the joints at each instant

Control	What can change	What should persist
Pose sequence	Body motion	Performer appearance
Camera poses	Viewpoint	Scene identity
Object trajectories	Where objects move	Object appearance

Chan et al. (2019); Hu et al. (2024); Wang et al. (2024)

Everybody Dance Now: Transferring Motion

Application: transfer the motion from one video to another video

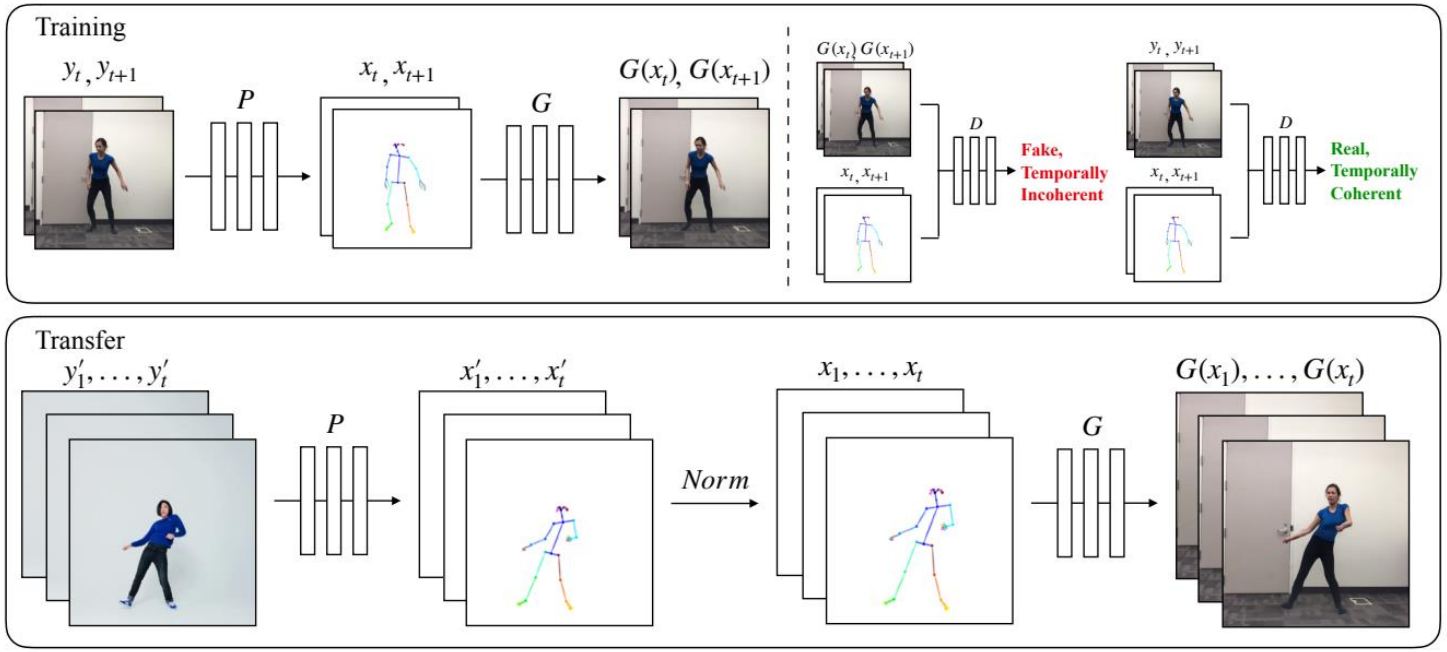


Chan et al., Everybody Dance Now (ICCV 2019) · Fig. 3; §3

Everybody Dance Now: Transferring Motion

Key idea: Use GAN to distinguish if two images are from the same person

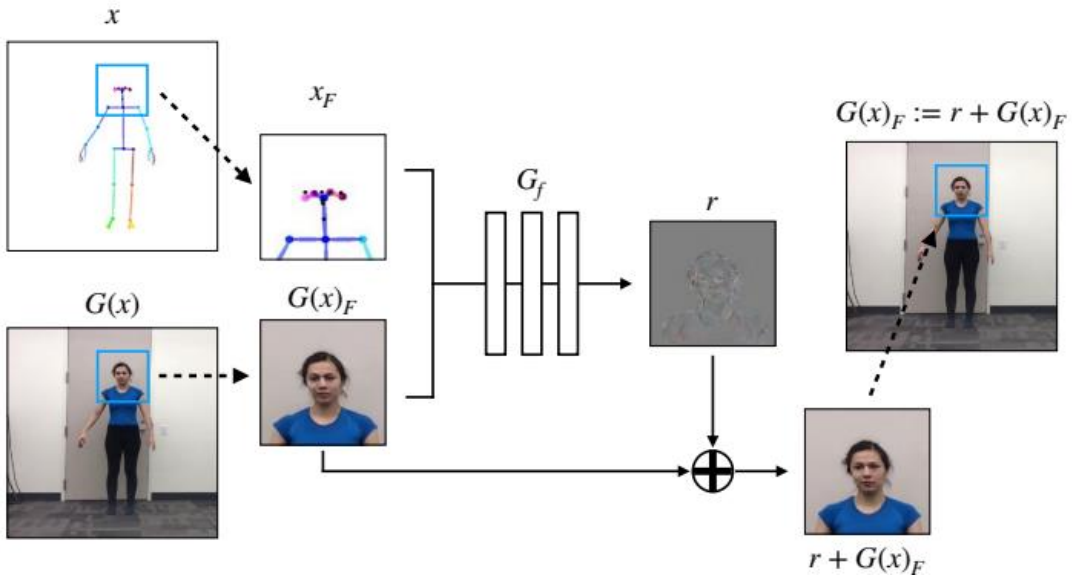
Then, at inference time, use different pose inputs



Chan et al., Everybody Dance Now (ICCV 2019) · Fig. 3; §3

Everybody Dance Now: coherence and its limits

Optional detail



Add a FaceGAN to ensure the reconstruction of face details

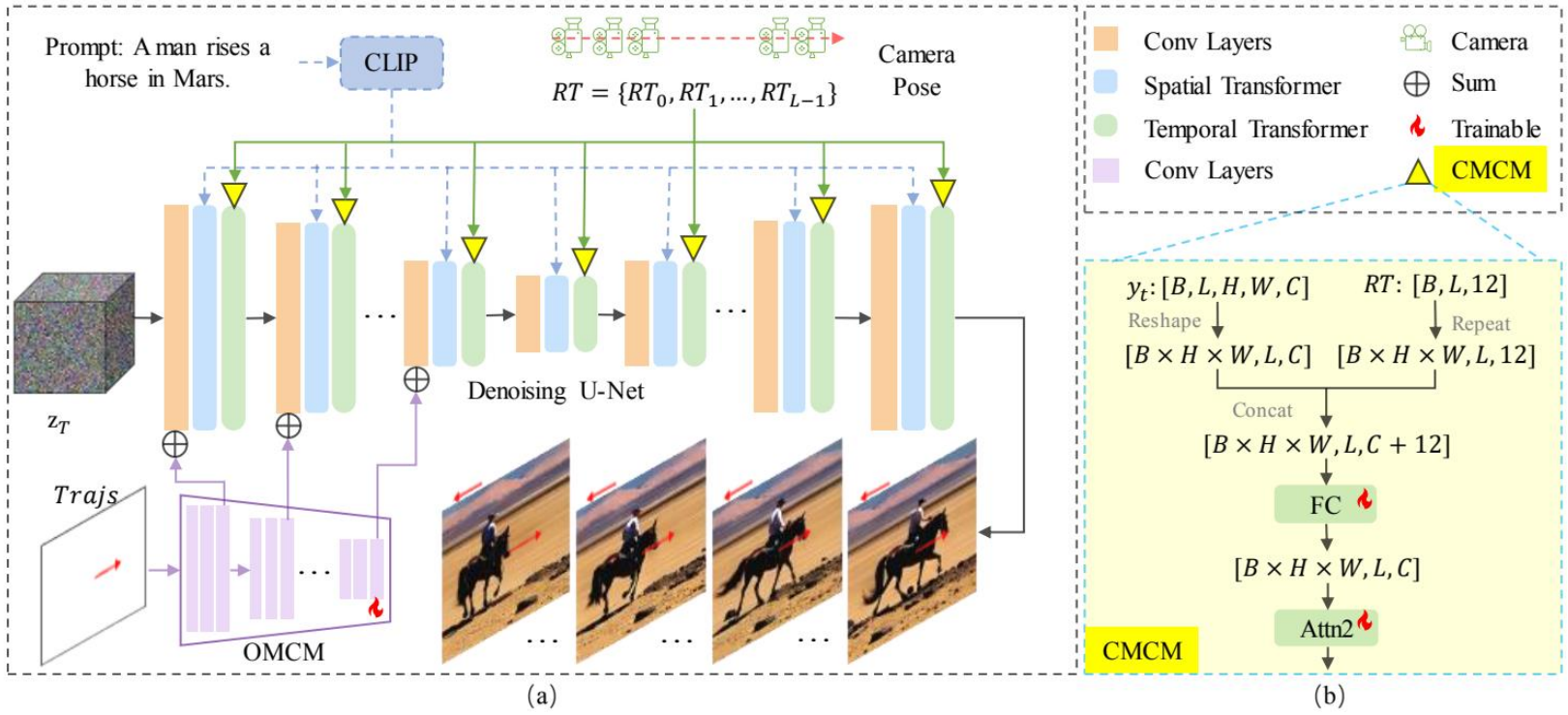
83.3% preferred the full method over PoseWarp in the paper's user study

PoseWarp uses a flow-prediction method to do keypoint-to-image synthesis

Chan et al., Everybody Dance Now (ICCV 2019) · Fig. 4; §3.2–4.2; §6

MotionCtrl: camera and object motion

Key idea: camera poses and object trajectories need separate control paths



Camera poses enter temporal attention; trajectory features enter spatial layers

Wang et al., MotionCtrl (SIGGRAPH 2024) · Fig. 2; §3.2

MotionCtrl: supervision and control accuracy

Optional detail

Camera control uses RE10K poses and generated captions

Object control uses WebVid tracks, learned densely and then sparsely

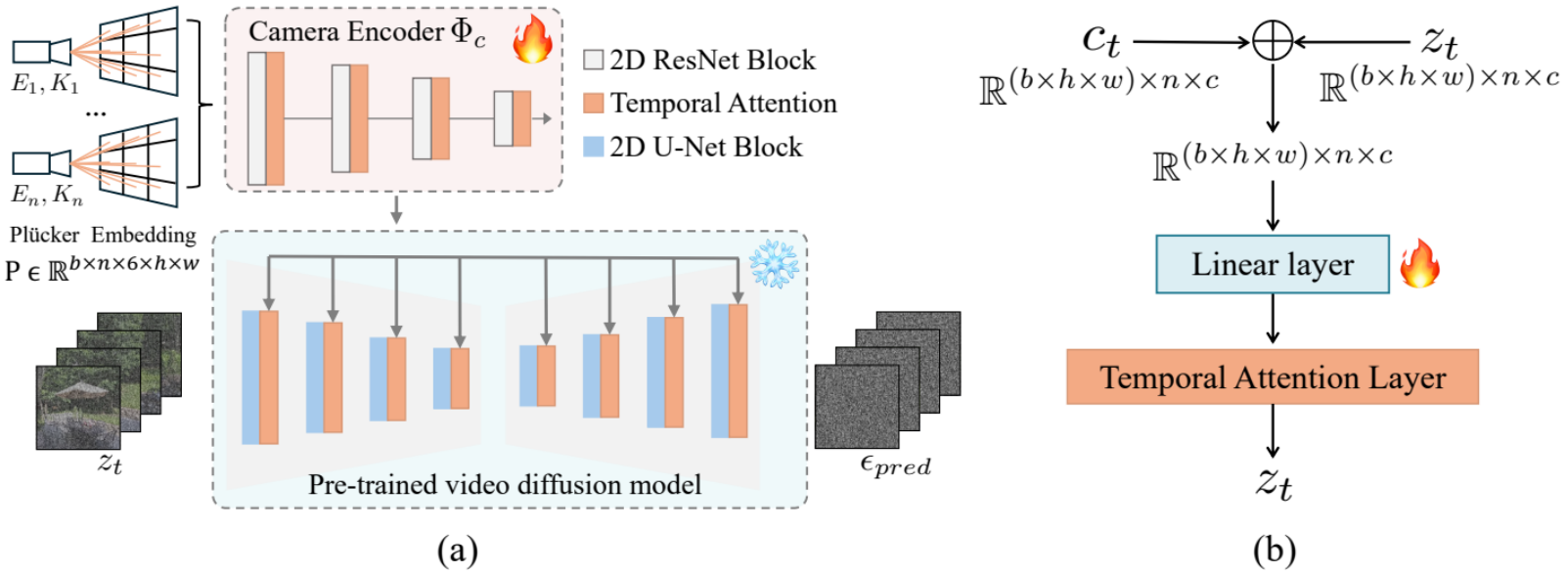
Motion error ↓	**VideoComposer**	**MotionCtrl**
Complex camera poses	0.0950	0.0735
Object trajectories	36.8351	28.877

Again, uses different models/datasets to generate training data

Wang et al., MotionCtrl (SIGGRAPH 2024) · §3.2.3; §4.1; Table 1; §5

CameraCtrl: camera motion as viewing rays

Key idea: give camera parameters a geometric meaning at every pixel

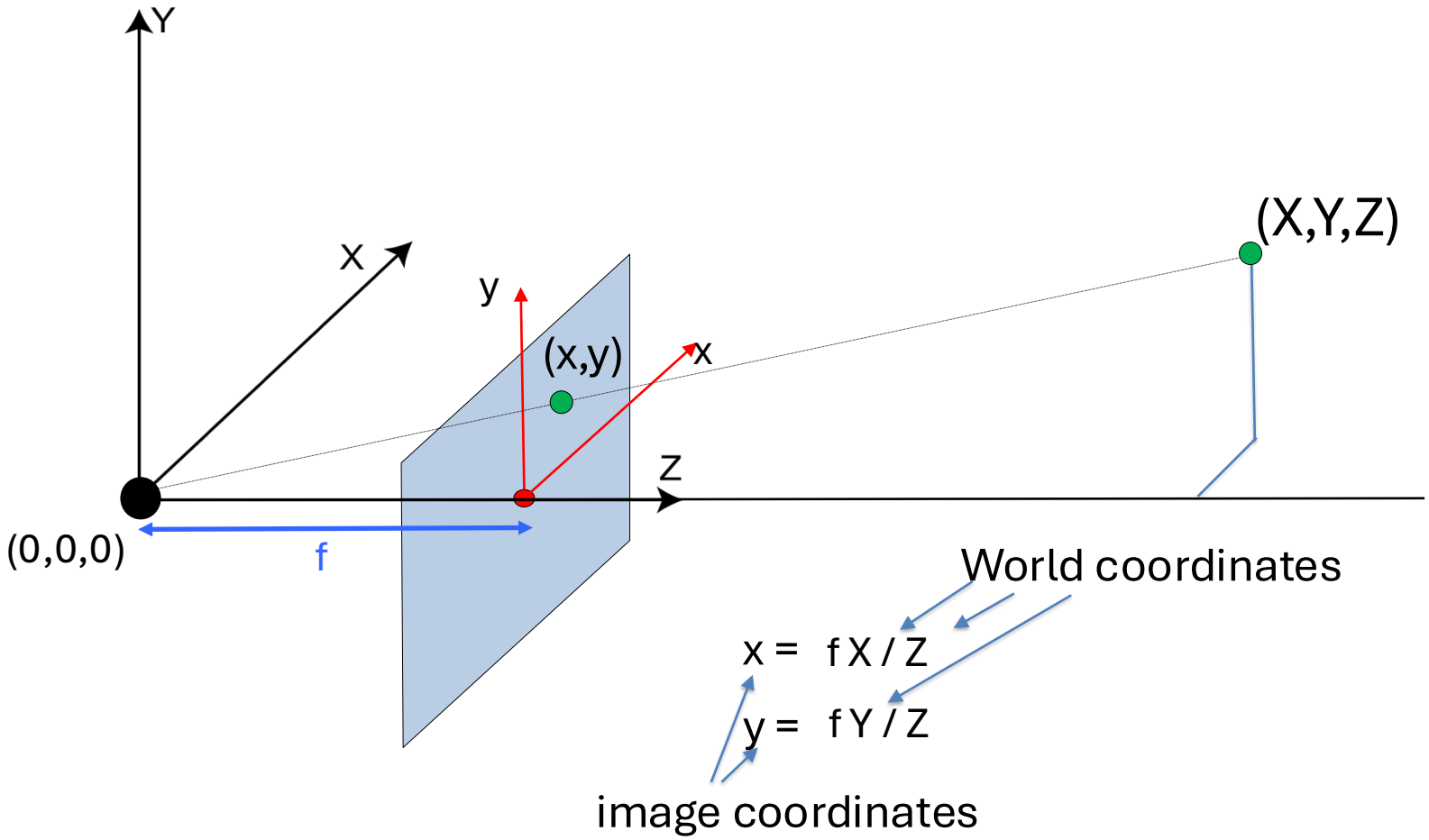


$$p_{u,v} = (o \times d_{u,v}, d_{u,v})$$

Each pixel contain information about the “ray” of the camera sensing

He et al., CameraCtrl (ICLR 2025), arXiv v2 · Fig. 2; §3.2–3.3

Real and Virtual Imaging Planes



If we know the camera facing direction, we know the direction of (X, Y, Z)

Summary: Ideas for (Conditional) Video Generation

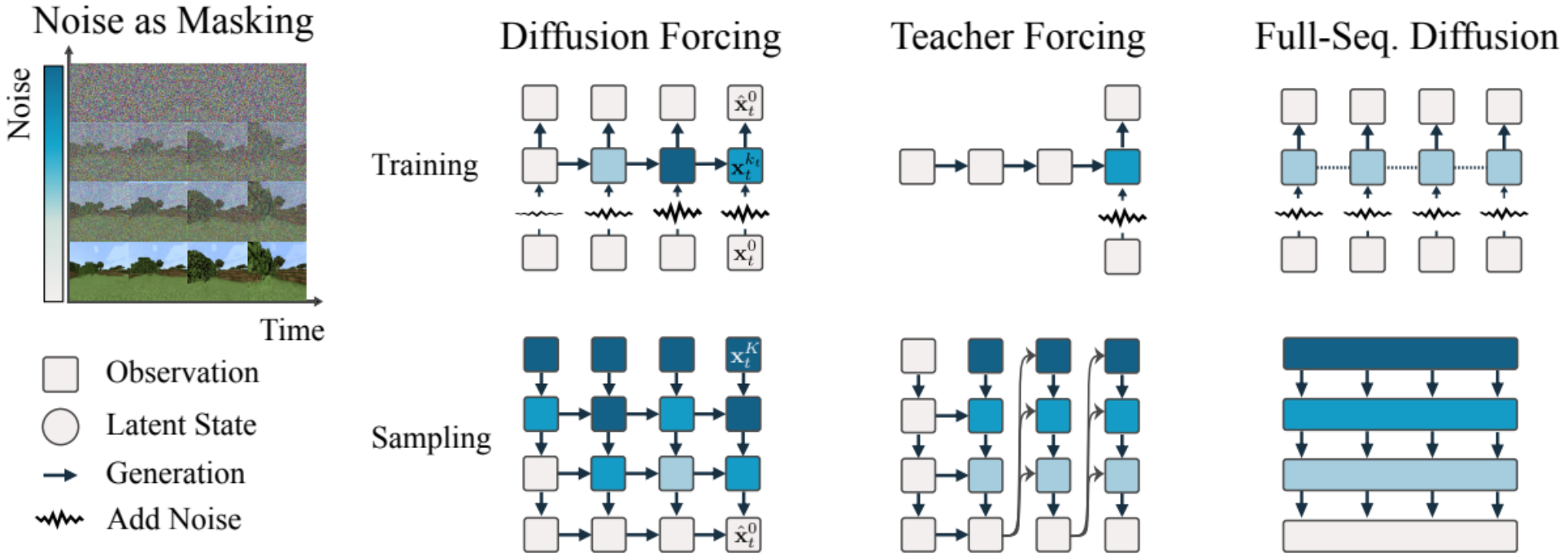
- VQ-VAE or 3D VAE codecs to compress
- Spatiotemporal global attention
- Spatiotemporal resolution hierarchies
- Image formation Ideas: Flow, 3D Rays
- Synthetically generated labels: camera motion, keypoints, etc.
- Specialized discriminators (for faces, texts, etc.)

4. Advanced Inference Techniques

Teacher Forcing and Full-Sequence Diffusion

Teacher forcing predicts the next token from a clean ground-truth history

Full-sequence diffusion jointly refines a target sequence



Chen et al., Diffusion Forcing (2024) · Fig. 2

Noise = “Soft” Masking

$$x_t^{k_t} = \alpha_{k_t} x_t + \sigma_{k_t} \varepsilon_t$$

Noise level	**Information retained**
Zero	A clean observed token
Intermediate	A partially informative token
Maximum	Approximately no original signal

Chen et al., Diffusion Forcing (2024) · §3.1

Training Time: Independent Noise Level Per Token

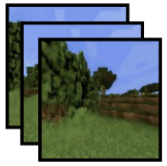
$$L_{DF} = E \sum_{t=1}^{T} \| \varepsilon_t - \varepsilon_\theta(h_{t-1}, x_t^{k_t}, k_t) \|^2$$

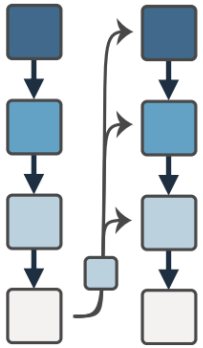
Sample a training sequence and an independent noise level for each token

Chen et al., Diffusion Forcing (2024) · Eq. 3.1; Algorithm 1

Inference Time: Different Ways to Set the Schedule

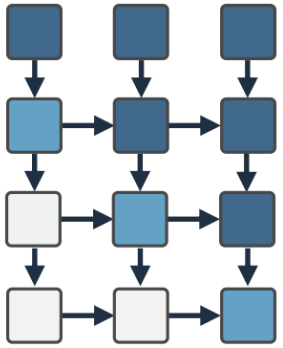
Stable Auto-Reg. Rollout





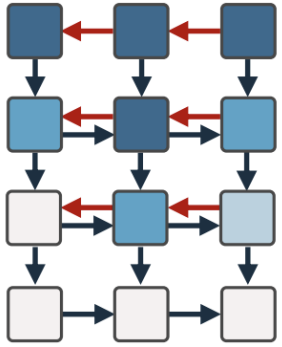
Diffuse w/ Causal Uncertainty





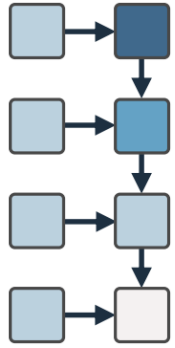
Full Traj. Guidance





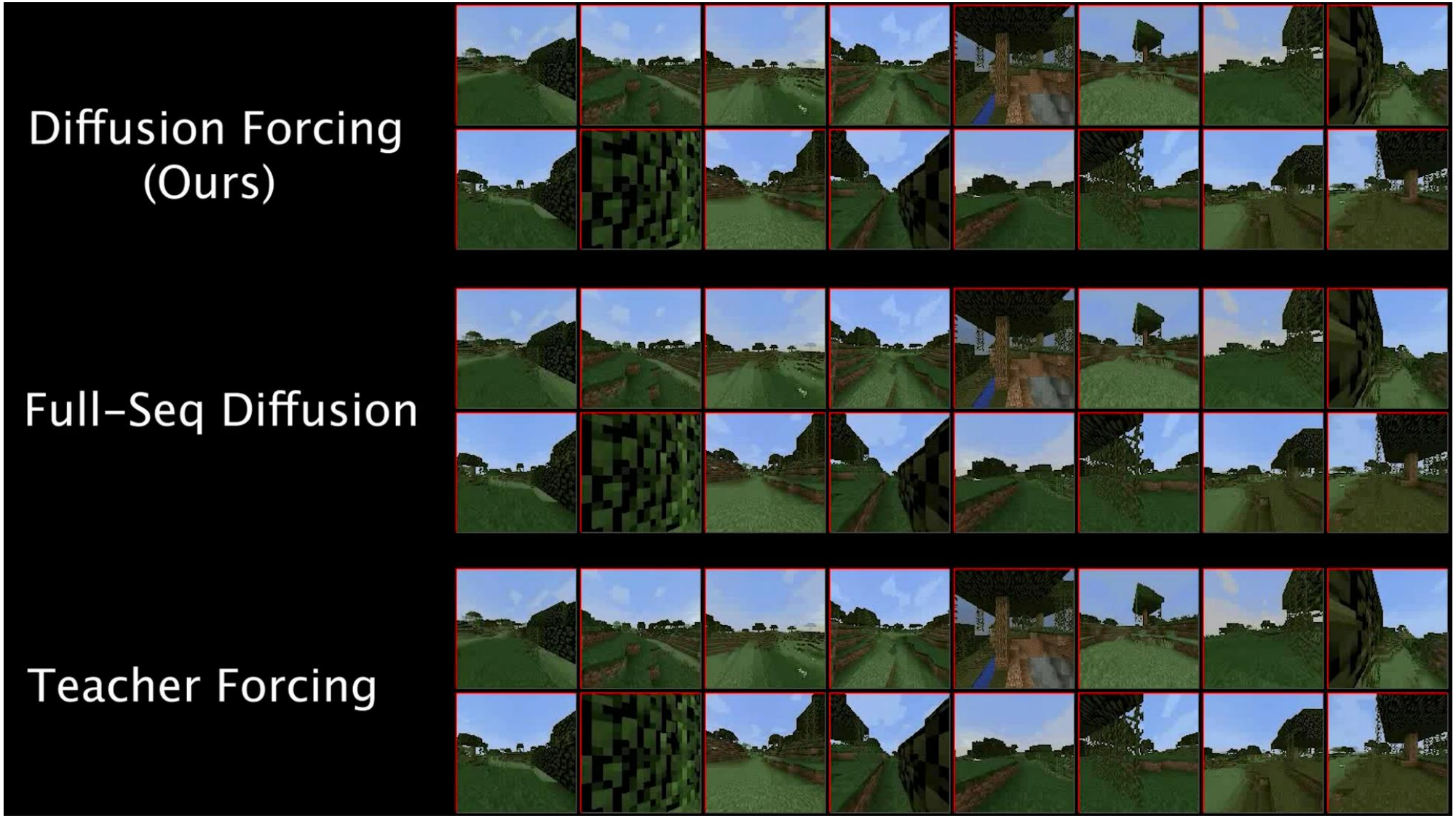
Condition on Currupted Obs.





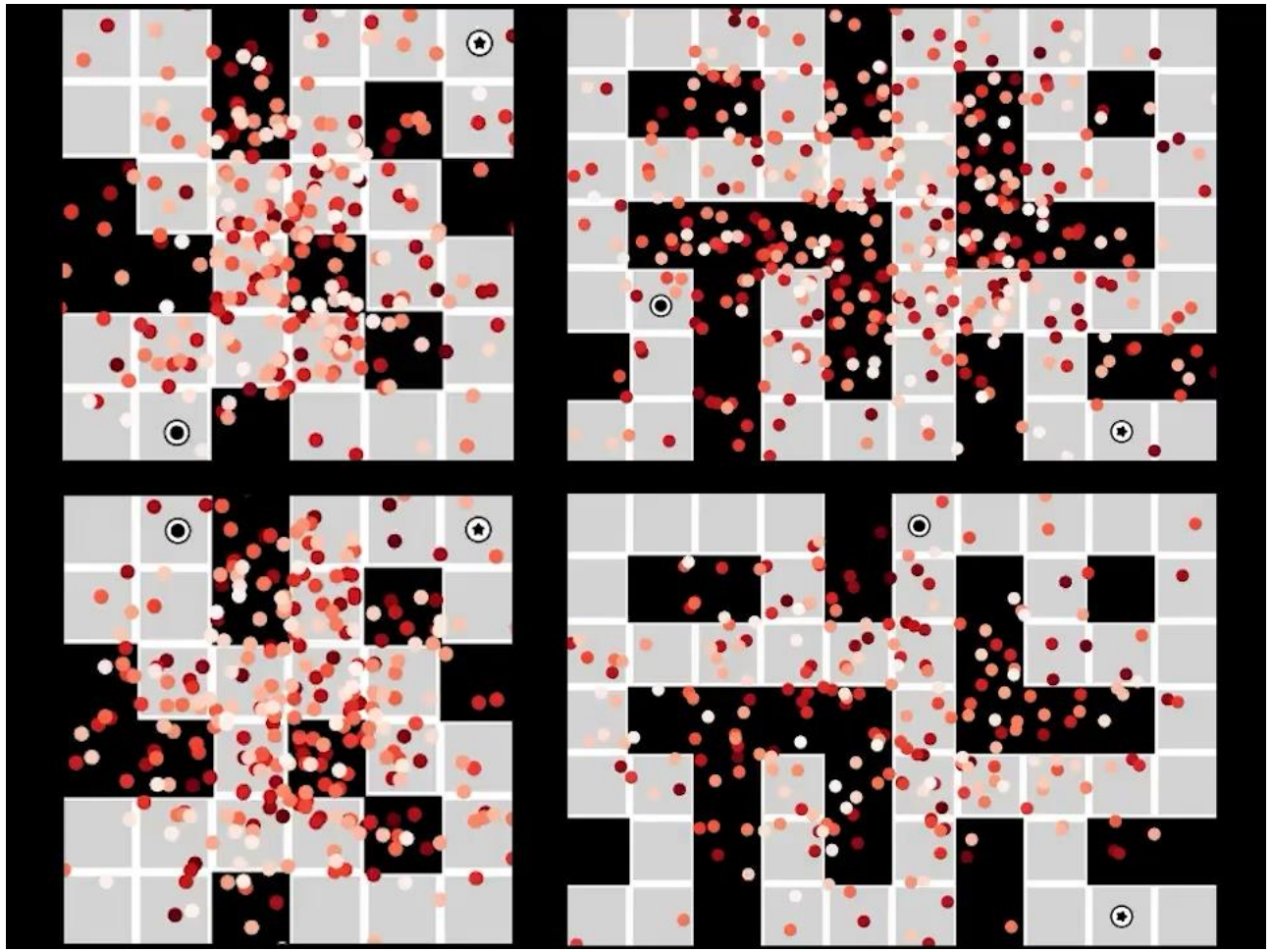
Chen et al., Diffusion Forcing (2024) · Eq. 3.1; Algorithm 1

Application 1: Improve Long-Horizon Prediction



Chen et al., Diffusion Forcing (2024) · Eq. 3.1; Algorithm 1

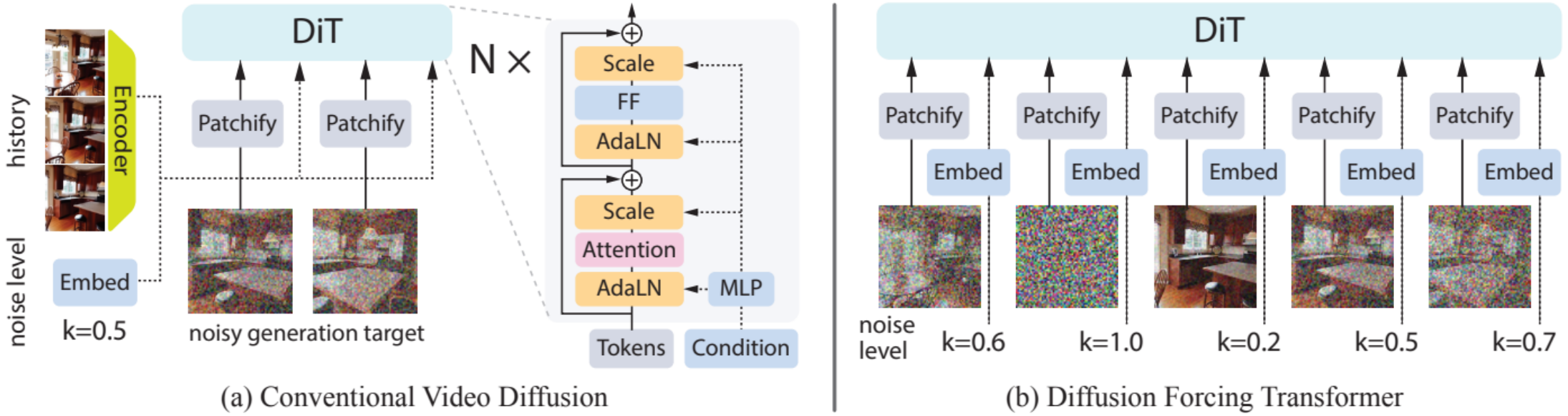
Application 2: Planning



Chen et al., Diffusion Forcing (2024) · Eq. 3.1; Algorithm 1

Application 3: Modeling History

Key idea: Use noise to supply information about “timestamps”



Song et al., History-Guided Video Diffusion (2025) · Fig. 2; §4

History Masking is Now Tunable

$$L_{DFoT} = E \|\varepsilon - \varepsilon_\theta(x^k, k)\|^2$$

Train with independently sampled noise levels for every frame

At inference, keep selected frames clean and denoise only the future

The same trained network supports prediction, interpolation, or mixed

Song et al., History-Guided Video Diffusion (2025) · Eq. 4; §4

Classifier-Free Guidance for History

$$s_{\text{guided}} = s_{\emptyset} + \omega(s_{H} - s_{\emptyset})$$

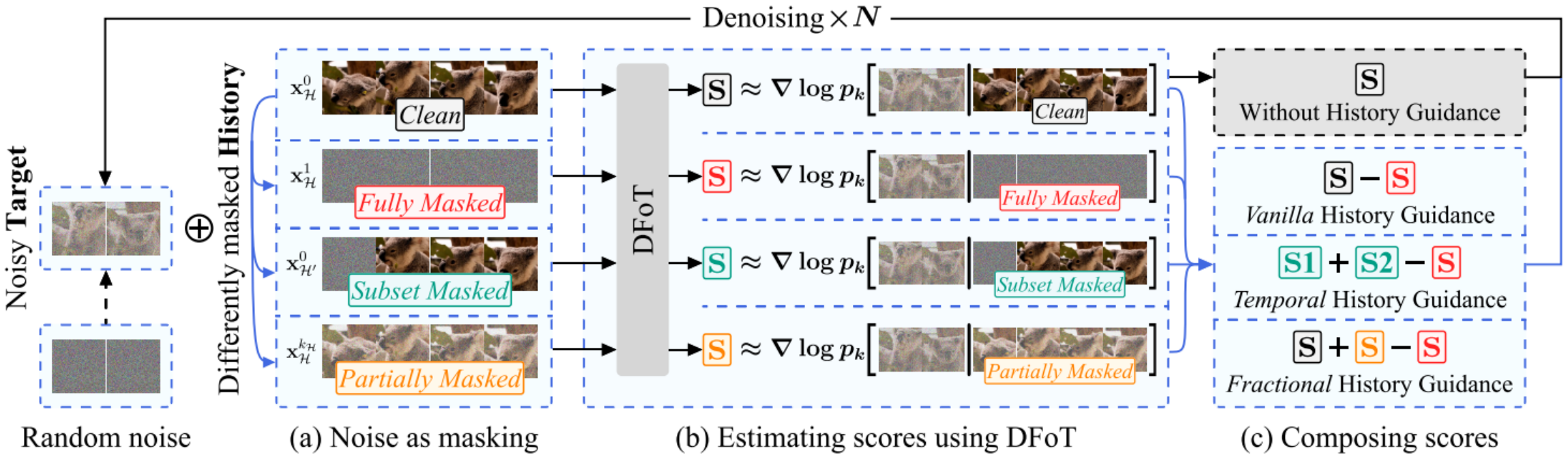
The conditional score uses clean history; the unconditional score masks that history with noise

A scale of one gives ordinary conditional sampling

Larger scales emphasize agreement with history

Song et al., History-Guided Video Diffusion (2025) · Eq. 2; §5

Classifier-Free Guidance for History: Different Strategies

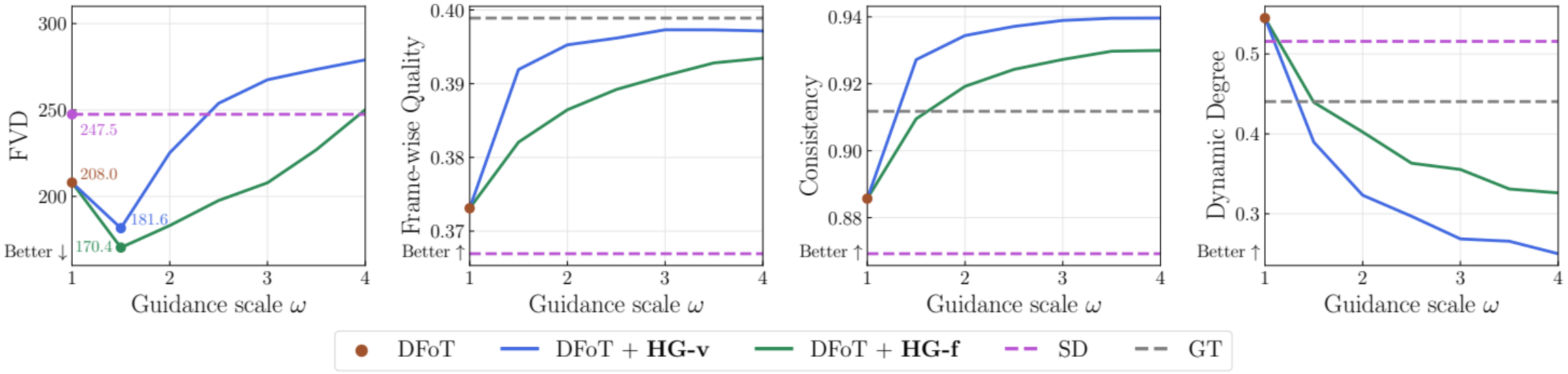


Song et al., History-Guided Video Diffusion (2025) · Fig. 3; §5

Guidance Parameter

Higher vanilla guidance improves visual quality and temporal consistency

But, it may reduce dynamics and diversity



Song et al., History-Guided Video Diffusion (2025) · Fig. 5; §6.3

Summary: Lverage the Noise Level in Diffusion

- Noise = “Soft Masking”
- This view unifies sequential prediction, sequence interpolation, and full “parallel” diffusion/prediction

Questions for Discussion

We have covered a lot of spatiotemporal data formats. For controllable generation, what would be useful conditional signals? How can you annotate them? How would you do evaluation?

Audio and video are both temporal data: what are the similarities/differences between them? How do these change our way of processing them?